

Inferential Feature Analysis: Latino Trump Support (CMPS 2020)

Analysis Pipeline — Random Forest + SHAP + Bootstrap

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Table 1: Analysis Configuration.

Parameter	Value
Bootstrap iterations (B)	500
Trees per forest	500
Test proportion	0.25
Rank stability threshold (k)	3
Development mode	FALSE
Raw data file	39096-0001-Data.rda

Table 2: Latino Subsample Summary.

Item	Value
Latino N	2220
Trump voters	585
Trump %	26.4%
Weight range	[0.14, 3.69]

15 Save Results Bundle

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16 Session Information

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Table 3: Train/Test Split Summary.

Set	N	Trump %	Features
Training	1665	26.1	2082
Test	555	27.0	2082

Table 4: Progressive Exclusion Tiers.

Tier	Train N	Test N	Features	Excluded
1: Full	1665	555	2082	0
2: Non-Tautological	1665	555	2022	60
3: Non-Partisan (Ring 1)	1665	555	1945	137

Table 5: Tier 3 AUC Across Learners.

Learner	Test AUC
Random Forest	0.965
Elastic Net Logistic	0.967
XGBoost	0.954

1 Analysis Parameters

2 Data and Sample Construction

3 Preprocessing

4 Train/Test Split and Imputation

5 Tier Construction and Partisan Leakage Audit

6 Model Fitting

7 Benchmark Models

Elastic net logistic regression and XGBoost are fitted on the same Tier 3 train/test split to confirm that findings are a property of the data rather than an artifact of Random Forest.

8 SHAP Computation

9 Feature Correlation Structure

When features are correlated, SHAP distributes predictive credit across correlated items in ways that are mathematically principled but may not map onto standalone constructs. The correlation matrix below documents the pairwise relationships among the top features.

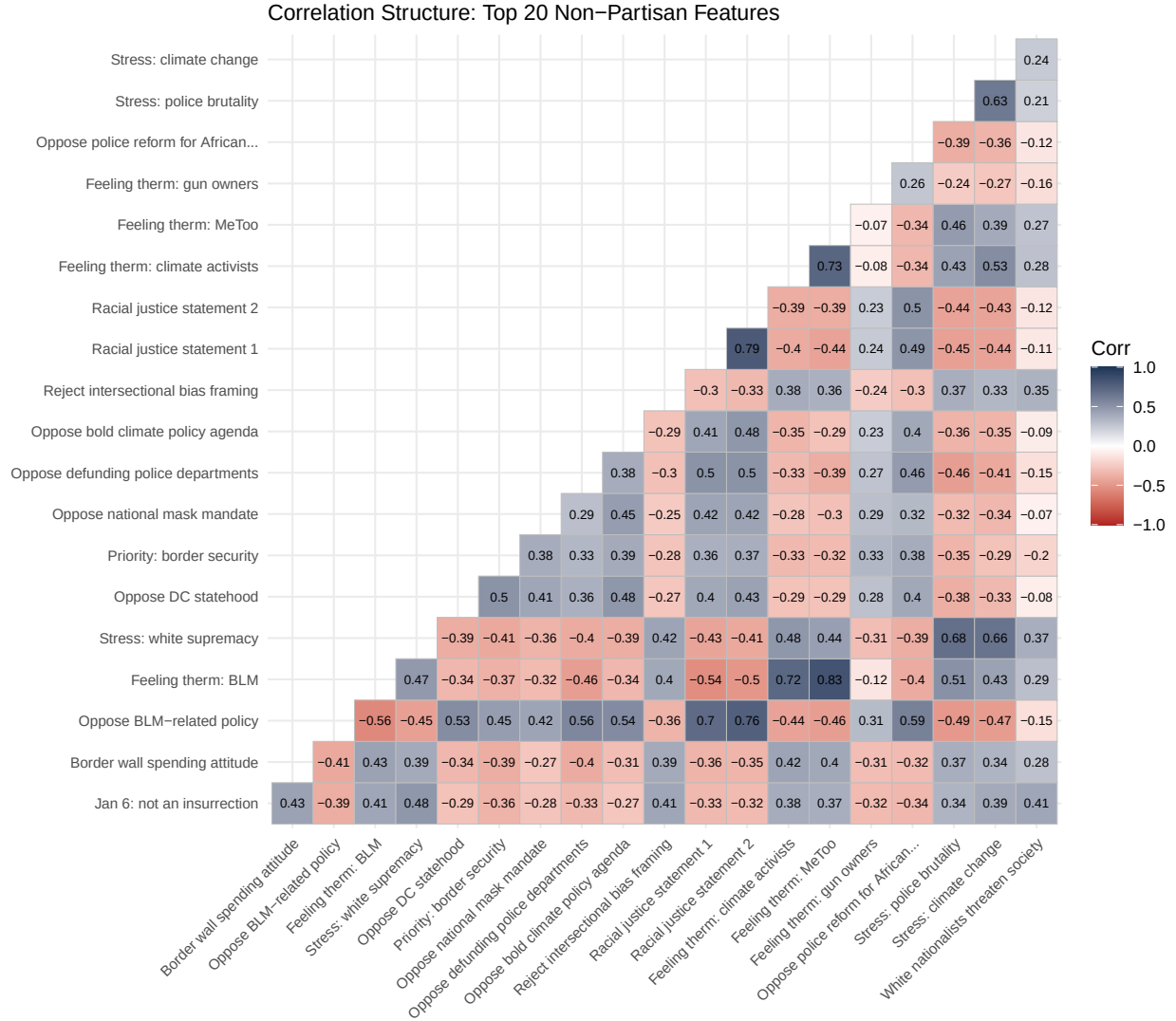


Figure 1: Pairwise Pearson correlations among top 20 Tier 3 features (test set).

10 Bootstrap Confidence Intervals

The bootstrap resamples the training set 500 times, refitting a Random Forest and computing TreeSHAP on the fixed test set each time. This isolates model instability from evaluation-set composition variance. Each feature's 95% CI is the 2.5th–97.5th percentile of the bootstrap distribution of $E_w[|\phi_j|]$.

Table 6: Threshold Sensitivity: Stable Features at $k = 2.0, 3.0$, and 4.0 .

Rank	Feature	sigma_j	k=2.0	k=3.0	k=4.0
1	Jan 6: not an insurrection	0.81			
2	Border wall spending attitude	1.69			
3	Oppose BLM-related policy	1.56			
4	Feeling therm: BLM	1.30			
5	Stress: white supremacy	2.34			
6	Oppose DC statehood	3.06			
7	Priority: border security	3.92			
8	Oppose national mask mandate	3.30			
9	Oppose defunding police departments	5.38			
10	Oppose bold climate policy agenda	4.33			
11	Reject intersectional bias framing	5.76			
12	Racial justice statement 1	4.77			
13	Racial justice statement 2	4.86			
14	Feeling therm: climate activists	4.63			
15	Feeling therm: MeToo	5.53			
16	Feeling therm: gun owners	4.43			
17	Oppose police reform for African Americans	5.45			
18	Stress: police brutality	4.92			
19	Stress: climate change	5.16			
20	White nationalists threaten society	7.80			

Table 7: Spearman Rank Correlations Across 5 Independent Splits (min = 0.7, mean = 0.714).

Split	Split_1	Split_2	Split_3	Split_4	Split_5
Split_1	1.000	0.716	0.717	0.714	0.710
Split_2	0.716	1.000	0.709	0.722	0.706
Split_3	0.717	0.709	1.000	0.723	0.724
Split_4	0.714	0.722	0.723	1.000	0.700
Split_5	0.710	0.706	0.724	0.700	1.000

11 Results Assembly

12 Threshold Sensitivity

The stability threshold k is a pre-specified reporting rule, not a universal standard. The table below shows which features qualify as stable findings at $k = 2.0$, $k = 3.0$, and $k = 4.0$, holding the magnitude criterion ($ci_lo > 0$) constant throughout.

13 Repeated-Resampling Robustness

The bootstrap assesses training-sample sensitivity conditional on one fixed test partition. To confirm the feature architecture is not an artifact of the particular split, five independent 75/25 partitions are drawn and the full Tier 3 RF + SHAP pipeline is run on each. Spearman rank correlations of feature importance across all pairs test whether the same features surface regardless of the split.

Table 8: AUC Across the Progressive Exclusion Framework.

Model	Test AUC	Drop from T1	Boot Mean AUC	95% CI
Tier 1 (Full)	0.992	NA	NA	—
Tier 2 (Non-Tautological)	0.980	0.013	NA	—
Tier 3 (Non-Partisan)	0.965	0.027	0.963	[0.958, 0.966]

Table 9: Stable Ring 1 Non-Partisan Features ($ci_lo > 0$, $\sigma_j \leq 3$).

Rank	Feature	Code	$E_w[\phi_j]$	CI lo	CI hi	σ_j
1	Jan 6: not an insurrection	Q53_2..A.coordinated.act.of.insurrection.against.the.United.States	0.01625	0.01171	0.02015	0.81
2	Border wall spending attitude	Q543_2..Oppose.government.increasing.spending.by..25.billion.on.border.security..including.building.a.wall.between.the.U.S.and	0.01316	0.00866	0.01708	1.69
3	Oppose BLM-related policy	Q358_5..Strongly.oppose	0.01294	0.00877	0.01605	1.56
4	Feeling therm: BLM	Q308R7	0.01267	0.00992	0.01670	1.30
5	Stress: white supremacy	Q504R6	0.00970	0.00698	0.01376	2.34

14 Main Results

14.1 Model Performance

14.2 Stable Non-Partisan Features

Features are reported as stable findings under a conjunctive decision rule pre-specified before examining results: the bootstrap 95% CI lower bound for $E_w[|\phi_j|]$ must exceed zero (magnitude criterion) AND $\sigma_j \leq 3$ (rank stability criterion). The first condition ensures the feature’s importance is bounded away from nothing across 500 training resamples. The second ensures its ordinal position in the importance ranking is stable enough to support comparative claims.

14.3 Figure 1. Feature Importance (Beeswarm)

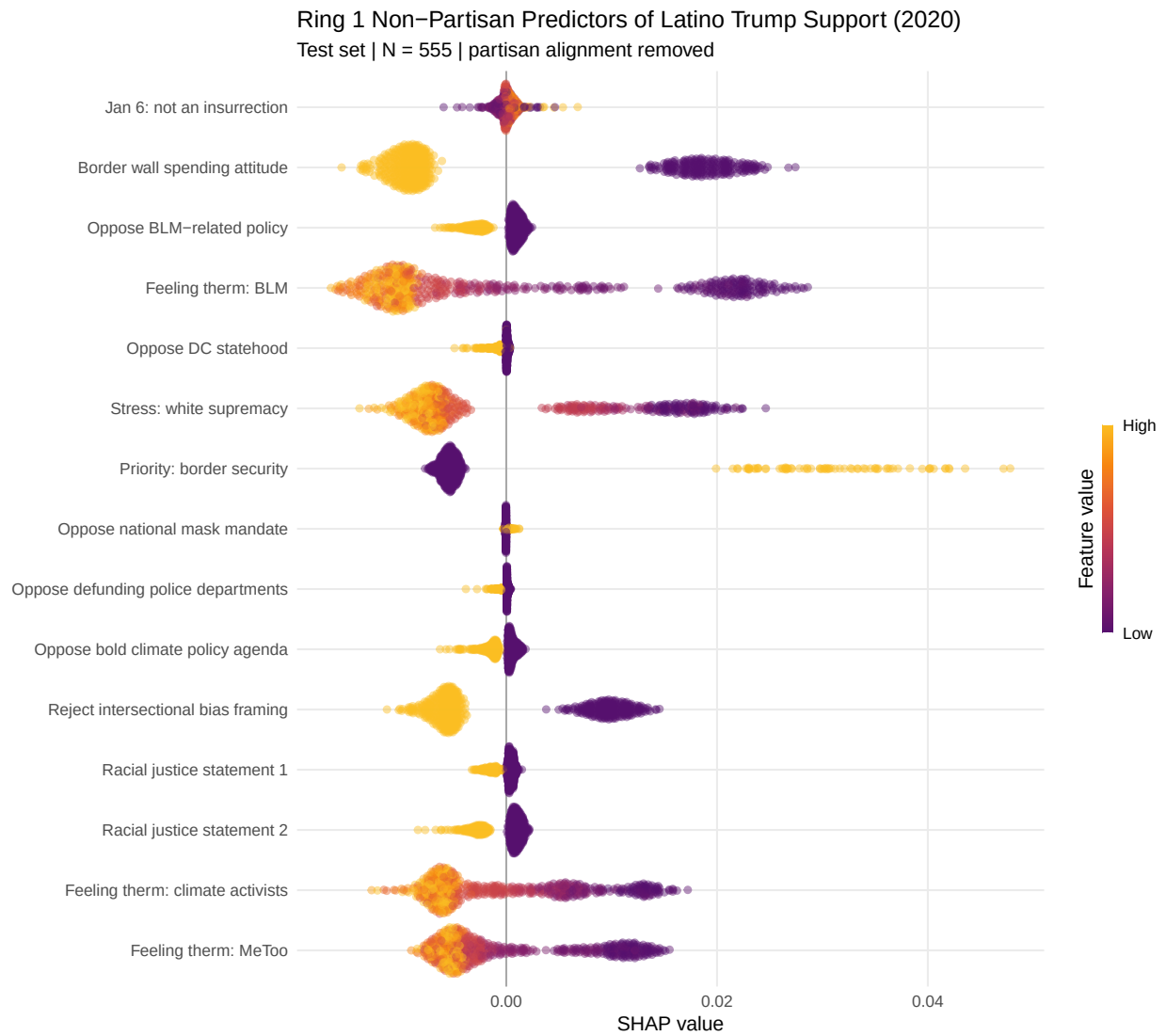


Figure 2

14.4 Figure 2. Bootstrap Confidence Intervals

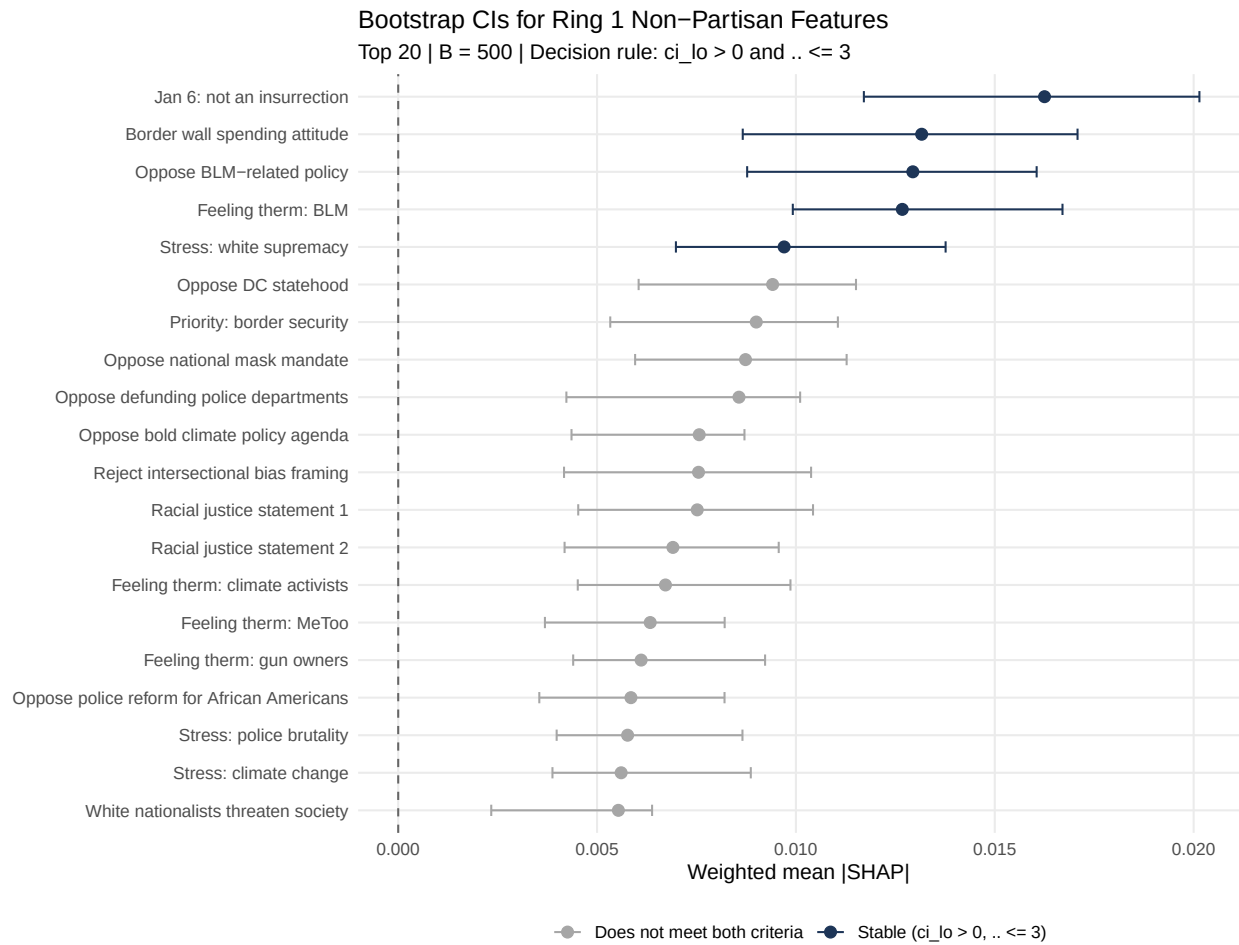


Figure 3

14.5 Figure 3. SHAP Dependence (Top Feature)

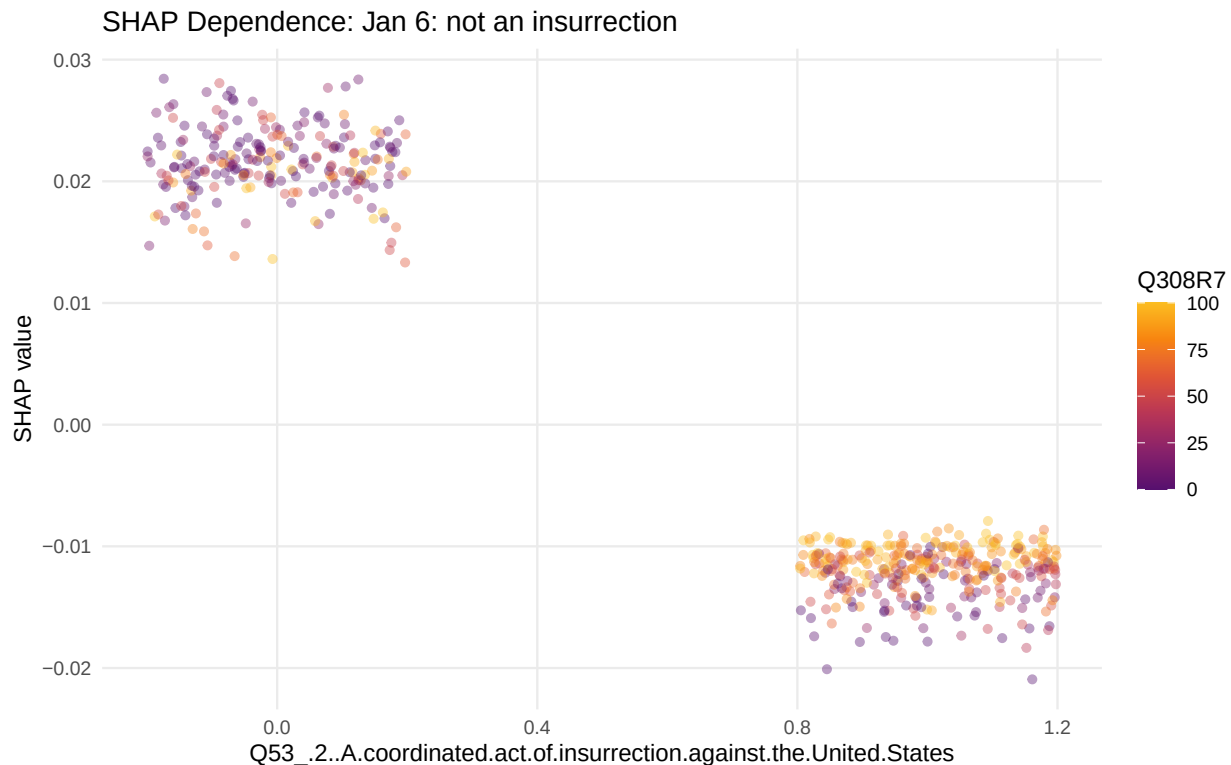


Figure 4

15 Save Results Bundle

All model objects, SHAP matrices, bootstrap results, tier metadata, and summary tables are saved to a single `.rds` file. The reporting QMD loads this file to generate all manuscript figures and tables without re-running the model.

The results bundle has been saved to `data/derived/ifa_results_2020.rds` (58 MB). The pooled paper loads this file with `readRDS("data/derived/ifa_results_2020.rds")`.

16 Session Information

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- Session info -----
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version  R version 4.5.3 (2026-03-11)
os       macOS Tahoe 26.5.2
system   aarch64, darwin20
ui       X11
language (EN)
collate  C.UTF-8
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ctype    C.UTF-8
tz        America/Indiana/Indianapolis
date      2026-07-14
pandoc    NA (via rmarkdown)
quarto    1.9.38 @ /Applications/RStudio.app/Contents/Resources/app/quarto/bin/quarto

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- Packages -----
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[2] /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library

* -- Packages attached to the search path.
